

Blackholes in the Milky Way?

A research team at Keio University has discovered intermediate-mass black hole (IMBH) at the centre of the Milky Way galaxy. <http://dgit.in/QekuTB>

Ghost galaxies discovered

Galaxies alleged to be fossils of the early universe, which haven't changed in 13 billion years have been unmasked by the Hubble Telescope

planet. Numerous books were written on the subject of extra-terrestrial life, Mars in particular. People such as Percival Lowell started publishing maps, being the owner of an observatory, he imagined he saw whole networks of canals and even went about labelling and charting them.

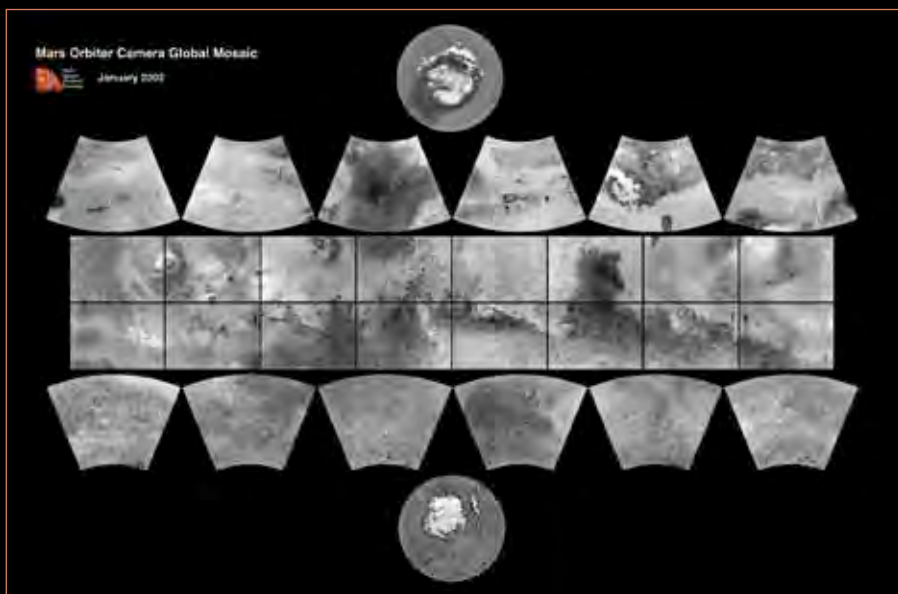
19th-20th Century - First Contact!

This 19th century saw the rise of the industrial revolution, the development of rocket engines and by the 20th century, the launch of the first satellites into space. Interestingly, it was the Russians who sent the first probes to Mars, the Mars-1, Zond 2 and Zond 3 probes, between 1962 and 1965. Only the Zond 3 managed to transmit images though. These were soon followed by probes Mars 3-7, and the Mariner series of probes from the US. It wasn't long before actual vehicles and probes landed on the Martian surface, the Viking 1 and 2 probes. These never did find any trace of life, but they did transmit the first images from the surface of the red planet.

The 20th century also saw the launch of the Hubble Space Telescope, which completely revolutionises space photography as we know it (wouldn't you love to play with a camera, which has a lens with a 7 ft aperture and a 180 ft focal length?). Many other "rovers", such as Spirit and Opportunity, have landed on Mars and have collected a lot of data with regards to the surface composition and atmosphere, and some of the most spectacular images of the red planet yet. Head over to <http://dgit.in/QcQual> for more information on these rovers.

Present day - A stark reality!

The Curiosity rover is the latest spacecraft that is scheduled to land on Mars. Its



primary goal, other than exploration, is to dig into the Martian soil and examine the samples for proteins and other compounds that can only indicate the presence of life. The red planet is a barren and desolate place, just the dust alone is enough to kill a normal human and completely clog machinery, unless adequately protected. It's possible that Mars did host the building blocks of life, but many scientists have concluded that it would not have been possible for Mars to host it for any definitive period of time. It is very likely that Mars had a lot of water earlier, but the fact is that Mars is so tiny and has been bombarded with so many meteors that it just couldn't hold on to its resources. The low mass of the planet means that every time there a reasonably large meteorite (about the size of a large house) smashes into the surface, Mars just loses another large chunk of its atmosphere and even mass to the black

depths of space. The lack of an active magnetic field also means that whatever remaining atmosphere is there, is regularly washed off by the solar winds. Our magnetic field actually protects us from the harsher reality of space, radiation from solar storms, gamma ray bursts, etc. Mars is just not up to the task, there is no way, with the currently available technology, that Mars can ever be a viable home to humans.

The future?

After all this, we really need to know whether we actually want to go to Mars. Yes, it is one of the closest planets to us and not as hostile as some other planets, but what purpose can it serve?

The planet cannot support life, the surface is subject to debilitating dust storms and the radiation alone would kill almost all organic life. The only reason NASA and other space agencies are looking at Mars now, is to find out if the building blocks of life originated on Mars and somehow made their way to Earth, and also, in effect, as a research project for future space missions. The future of Mars, as a habitable alternative to our own is suspect. The future of space travel? Not by a long shot. We've really come a long way, from the first ape that threw a stone at the moon a million years ago, to actually landing a rocket on it 50 years ago. We are capable of anything and we will find a way to conquer space itself. Just probably not in this lifetime! **d**

Phobos probes

Russian Phobos 1 probe was probably among the unluckiest probes to visit the red planet. It was lost when the termination signal for an altitude correction program was activated by an incorrectly coded program. The program shouldn't have been there in the first place, but a rushed schedule meant that it got neglected.

The Phobos 2 probe was the source of a great controversy. The probe functioned perfectly for a long time, but when it came to within 50 feet of the Martian surface, to dispatch two landers, all communication was lost. What started the controversy was the presence of shadow in the shape of an ellipse, on the Martian surface, on the last image transmitted by Phobos 2 just before its destruction! The Soviets claim that it was just the shadow of the probe itself, but conspiracy theorists beg to differ.